ANSWERS

I. 1. (c) 2. (b) 3. (a) 4. (a) 5. (d)

6. (c) 7. (d) 8. (*c*) 9. (*a*) 10. (a)

11. (a) 12.(a) 13. (c) 14. (b) 15. (a)

16. (c) 17. (a) 18. (b) 19. (d) 20. (b)

II. 34. reject H_0 35. 0.1835 36. one-tailed test (right)

37. $H_0: \mu = 30$, $H_1: \mu < 30$ 38. −1.645 39. −1.645

39. Parameter = Mean height of X standard students in the school = 150 *cms*

Statistic = Average height of sampled students in the school = 155 *cms*

40. $SE(\bar{X}) = 3$

41. $SE(p) = 0.05$

III. 47. $z_0 = -4$

48. 0.089

49. 0.8106

50. 4.122

51. 1.937

IV. 58. $z_0 = 2.5$; do not reject H_0

59. $z_0 = -1.75$; do not reject H_0

60. $z_0 = 0.619$; do not reject H_0

61. $z_0 = -7.04$; reject H_0

62. $z_0 = 0.9619$; do not reject H_0

63. $z_0 = -1.638$; do not reject H_0

64. $z_0 = 2.508$; do not reject H_0

65. $z_0 = -3.167$; do not reject H_0

66. $z_0 = 3.90$; reject H_0

67. $z_0 = 2.55$; reject H_0

68. $z_0 = 1.352$; do not reject H_0

69. $z_0 = -1.444$; do not reject H_0

70. $z_0 = -0.886$; do not reject H_0

12th_Statistics_EM_Unit_1.indd 35 1/28/2020 4:47:43 PM